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Weakened Interfacial Hydrogen Bond Connectivity Drives Selective Photocatalytic Water Oxidation toward H_2O_2 at Water/Brookite-TiO_2 Interface

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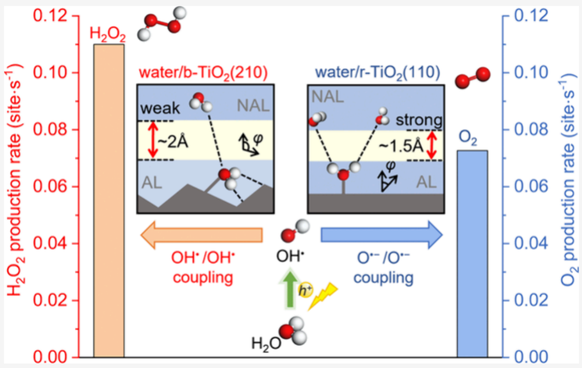

ABSTRACT: The formation of H_2O_2 through the two-electron photocatalytic water oxidation reaction (WOR) is significant but encounters the competition with the four-electron O_2 evolution reaction. Recent studies showed a crystal-phase dependence in H_2O_2 selectivity, where high purity brookite TiO_2 (b-TiO_2) exhibits remarkable H_2O_2 selectivity in contrast to the common rutile phase TiO_2 (r-TiO_2). However, the origin of such a structure-induced selectivity preference remains elusive, primarily due to the complexities associated with the solid−liquid interface system and excited-state chemistry. Herein, we conducted a comprehensive investigation into the selectivity mechanism of WOR at the water/b-TiO_2(210) and water/r-TiO_2(110) interfaces, employing first-principles molecular dynamics simulations and micro-kinetic analyses. Intriguingly, our results reveal that the intrinsic catalytic ability of the b-TiO_2(210) itself does not enhance H_2O_2 selectivity compared to r-TiO_2(110). Instead, it is the weakened interfacial hydrogen bond connectivity, modulated by the herringbone-like local atomic structure of the b-TiO_2(210) surface, that determines the selectivity. Specifically, this weakened H-bond connectivity (i.e., local low water density) at the interface, owing to the strong water adsorption and distinct adsorption orientation, can stabilize the $OH^•$ radical and inhibit its deprotonation, leading to an improved H_2O_2 selectivity. By contrast, the relatively strong interface H-bond connectivity established over r-TiO_2(110) accelerates the deprotonation of $OH^•$, with the $OH^•$ coverage being 3 orders of magnitude lower than at the water/b-TiO_2(210) interface. This study quantitatively demonstrates that the local H-bond structure (water density) at the liquid/solid interface significantly influences photocatalytic selectivity, and this insight may offer a rational approach to enhance the H_2O_2 selectivity.

■ INTRODUCTION

The photocatalytic water oxidation reaction (WOR) to produce the high-value-added chemical H_2O_2, via a two-electron pathway ($2H_2O + 2h^+ \rightarrow H_2O_2 + 2H^+$), has received increasing attention.[1−3] However, it faces competition with the four-electron oxygen evolution process ($2H_2O + 4h^+ \rightarrow O_2 + 4H^+$) in practical application,[4−6] despite the two-electron pathway requiring fewer electrons. Achieving a high H_2O_2 selectivity in this process has proven challenging, particularly for the widely used rutile phase TiO_2-based materials (r-TiO_2).[6−8] Despite great efforts to improve the H_2O_2 selectivity through various methods, such as doping, surface modification, and the development of new catalysts,[9−14] progress has fallen far short of expectations. Interestingly, it has recently been shown by simple crystal phase control that brookite-phase TiO_2 (b-TiO_2) with high purity exhibits remarkable selectivity for H_2O_2 generation.[4,15,16] This is particularly notable on the b-TiO_2(210) surface, which has a high rate of $OH^•$ radicals, a key intermediate for H_2O_2 generation.[7,17,18] This selectivity has not been observed on the extensively studied r-TiO_2(110) surface.[7,19] Considering the configuration of both crystal phases, which share the same composition and similar atomic coordination configurations (Scheme 1) and expose 5-fold coordinated surface Ti sites (Ti_{5c}, a key active site for WOR), they may have a similar intrinsic catalytic ability. This raises the question: what is the key factor determining the reaction selectivity on different crystal phase surfaces?

Understanding the atomic-level mechanism of WOR is essential for elucidating the crystal-phase dependence and strategically enhancing the selectivity of H_2O_2. However, current studies face challenges in gaining the mechanistic insight. First, WOR involves multiple competing pathways with complex intermediates. Notably, the formation of peroxide intermediates encompasses diverse mechanisms, such as the oxo−oxo coupling of $OH^•$ radicals or $O^{•-}$ radicals on two

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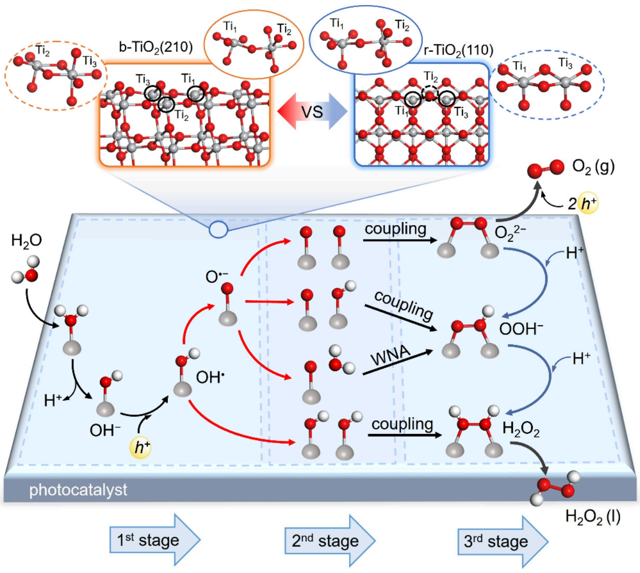
Scheme 1. Coordination Configurations of b-TiO_2(210) and r-TiO_2(110) and the General Mechanism for Photocatalytic WOR into H_2O_2/O_2 at Three Key Stages[a]



[a]1st stage: O−H bond dissociation. 2nd stage: O−O bond formation. 3rd stage: O_2/H_2O_2 evolution. Gray, Ti; red, O; white, H.

metal sites,[6,20,21] alongside the water nucleophilic attack (WNA) on a M−O$^\bullet$ species.[21−23] Subsequently, the conversion of peroxide intermediates may lead to the formation of H_2O_2 through protonation or the production of O_2 via deprotonation.[13,17] The intricate nature of these competing reaction pathways poses a formidable challenge in discerning the mechanisms and kinetics governing H_2O_2 and O_2 evolution. Second, the photocatalytic WOR involves intermediates in excited states, including OH$^\bullet$ radicals, O$^{\bullet-}$ radical anions, and peroxide intermediates. Explicitly simulating these photoinduced intermediates and reaction on the catalyst surface remains a difficult task.[6,24,25] Third, apart from the photoexcitation condition, the solid/liquid interface introduces a complex hydrogen-bond network and a dynamically fluctuating aqueous environment. Few studies have comprehensively addressed the impacts of these reaction environments on surface radicals and their conversion. Notably, various liquid/solid interfacial environments including microdroplets,[26,27] poor wettable surface,[28] and hydrophobic surfaces[29,30] have demonstrated significant influences on the formation of H_2O_2. Therefore, a rigorous examination of the impact of reaction environment on the mechanism and kinetics of H_2O_2 and O_2 evolution is imperative.

In this study, the ab initio molecular dynamics (AIMD) simulations[6,31,32] and first-principles microkinetic analysis[33−35] were employed to systematically investigate the factors governing selective formation of H_2O_2 in the photocatalytic water oxidation reaction at the water/b-TiO_2(210) interface. Our aim was to elucidate the origin of the crystal phase dependence in determining the product selectivity in the presence of an explicit water solution. We analyzed the overall activity trend and identified the dominant reaction pathway, surface coverages of crucial intermediates, and the major kinetic obstacles. Comparative analysis with the r-TiO_2(110) catalyst revealed that b-TiO_2(210) exhibits similar intrinsic activity for OH$^\bullet$ radical formation. However, the crucial factor improving the selectivity of H_2O_2 is the weakened H-bond network connectivity at the interface, which is modulated by the local atomic structure on b-TiO_2(210). This interface environment significantly improves the selectivity by affecting the coverage of semihydrophobic OH$^\bullet$ and the oxo−oxo coupling behavior. This study represents an advancement in understanding the water/catalyst interface system at the atomic level, quantitatively demonstrating the controlling influence of H-bond network connectivity at the interface on the photocatalytic pathways and selectivity.

■ METHODS

DFT Calculations and Radical Species Simulation. All the spin-polarized calculations were performed with the Perdew−Burke−Ernzerhof (PBE) functional within the generalized gradient approximation (GGA)[36] using the Vienna ab initio simulation package (VASP, version 5.4.1).[37,38] The project augmented wave (PAW) method was used to represent the core−valence electron interaction,[39] and the valence electronic states (pseudopotential of H, O, Ti_pv in PBE_2010 version) were expanded in plane-wave basis sets with an energy cutoff of 450 eV. The PBE+U method was applied to run the molecular dynamics simulations as well as the structure optimization. U = 6.3 eV for O 2p orbitals and U = 4.2 eV for Ti 3d orbitals were chosen.[25,40] All the atoms were allowed to relax until the Hellman−Feynman forces on each ion were less than 0.05 eV/Å by using the quasi-Newton Broyden minimization scheme.[41] The van der Waals forces were considered in this work, using the PBE-D3 method with Becke−Jonson damping.[42,43] The constrained optimization scheme[44,45] was used to search the transition states (TS), and the distance of the atoms that will form new bond is constrained at an estimated value. The TSs can be located via changing the fixed distance and were verified when (i) all forces on the atoms vanish and (ii) the total energy is a maximum along the reaction coordinate but a minimum with respect to the rest of the degrees of freedom.

To simulate a photogenerated hole in the consideration of simplifying the complicated photoexcited system, we introduced an OH group as an electron acceptor on the bottom surface of the slab, resulting in a hole in the system. The localization of hole is confirmed by the electronic structural analysis, including the site-projected magnetic moment and Bader charge analysis.[6,24,25,40] The simulation details on the bulk hole and OH$^\bullet$ and O$^{\bullet-}$ radicals involved in WOR can be seen in the Supporting Information. It is worth noting that, in these strategies, the uncertain and relatively small electron−hole interaction that varies with the trapping center in the TiO_2 system was neglected while ensuring the electroneutrality of the periodic system (eliminating the energy correction resulting from the presence of background counter charge). It is worth noting that the HSE06 hybrid function (25% exchange mixing coefficient) has been used to optimize/check the properties of hole polaron[25,40] and the O−O bond formation barriers (Table S1), which serves as an illustration, affirming the rationality of our calculation settings.

Water/TiO_2 Interfaces and ab Initio Molecular Dynamics. We conducted a comprehensive investigation of all feasible pathways involved in the photocatalytic WOR to produce O_2 and H_2O_2 on the b-TiO_2(210) and r-TiO_2(110) surfaces in the presence of explicit water solution. Figure S1 illustrates the water/b-TiO_2(210) interface, consisting of a four-Ti-layer p(1 × 1) brookite(210) slab with 25 H_2O molecules confined by Ar atoms. A vacuum layer of 20 Å was applied along the z direction to avoid interactions between periodically repeated slabs. Similarly, the water/r-TiO_2(110) interface comprises a four-Ti-layer p(1 × 4) rutile(110) slab with ∼10 Å water, containing 26 H_2O molecules. To account for the dynamic water structure's influence on the reaction process, AIMD simulations were employed.[6,30] The canonical (NVT) ensemble with local Nosé−Hoover thermostats was used at T = 298 K with a 0.5 fs time step. Then, the reaction energetics were determined using the multipoint averaging molecular dynamics approach (MAP-MD).[6] For each intermediate, we selected ∼10 samples every 0.2 ps from the

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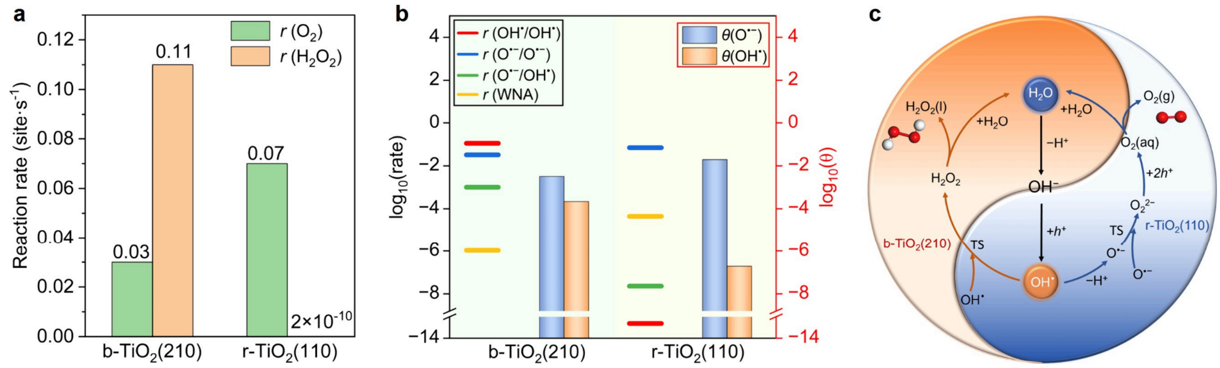

Figure 1. Kinetics analysis and proposed reaction mechanisms of WOR on the aqueous b-TiO_2(210) and r-TiO_2(110) surfaces. (a) Reaction rates (r) of O_2 and H_2O_2 at the water/b-TiO_2(210) and water/r-TiO_2(110) interfaces under the conditions $C_{hole} = 10^{-9}$ ML, $X_{H2O} = 1$, $X_{O2} = 10^{-7}$, pH = 7, T = 298 K. (b) Reaction rates of four possible mechanisms for O−O bond formation and the coverages (θ) of $O^{\bullet-}$ and $OH^{\bullet}$ intermediates. (c) The favored mechanisms of H_2O_2 and O_2 evolution at the water/b-TiO_2(210) and water/r-TiO_2(110) interfaces, respectively.

equilibrium structure of AIMD and further optimized them to obtain the total energy of each structure (E_{tot}). Since different water network configurations certainly affect E_{tot}, one needs to deduct the contribution of water solution (E_{water}) from E_{tot} but keep the solvation effect into consideration. Finally, we averaged the obtained solvation-like energies of all the samples in each intermediate and adopted the averaged value as the representative energy of each intermediate (see more details in the Supporting Information or ref 6). The corresponding energies for all reactions are given in Table S2. Such an approach has been used in our previous work, demonstrating high accuracy comparable to the state-of-art but time-consuming constrained molecular dynamics simulation.[6,31,32]

■ RESULTS AND DISCUSSION

Mechanism and Selectivity of WOR. Scheme 1 outlines the possible pathways involved in photocatalytic WOR.[6,13,17,21] In the first stage, the adsorbed water (H_2O_{ad}) on the Ti_{5c} site undergoes deprotonation, forming an OH^- species at the interface. Under photoirradiation, the OH^- species can be oxidized by a photohole, leading to the formation of $OH^{\bullet}$ radical, which aligns with the results of the previous results.[6,46] Next, the $OH^{\bullet}$ radical can transform into the $O^{\bullet-}$ radical anion on the surface by releasing a H^+ into the solution. In the second stage, the surface radical intermediates ($OH^{\bullet}$ and $O^{\bullet-}$) can couple together via oxo−oxo coupling with three possible pairs: (i) $OH^{\bullet}/OH^{\bullet}$; (ii) $O^{\bullet-}/OH^{\bullet}$; and (iii) $O^{\bullet-}/O^{\bullet-}$ (eqs 1−3), resulting in the generation of peroxide intermediates. For comparison, we considered the WNA mechanism,[22,23] involving the nucleophilic attack of water in the aqueous solution (H_2O_{aq}) to the $O^{\bullet-}$ radical (eq 4; see Figure S2).

$$OH^{\bullet} + OH^{\bullet} \rightarrow H_2O_2 \tag{1}$$

$$OH^{\bullet-} + OH^{\bullet} \rightarrow OOH^- \tag{2}$$

$$O^{\bullet-} + O^{\bullet-} \rightarrow O_2^{2-} \tag{3}$$

$$O^{\bullet-} + H_2O_{aq} + h^+ \rightarrow OOH^- + H^+(sol) \tag{4}$$

In the third stage, the peroxide intermediates produced via the O−O bond formation, including adsorbed H_2O_2, OOH^-, and O_2^{2-}, will transform into H_2O_2 or O_2. H_2O_2 can be produced through either a direct one-step protonation from OOH^- or an indirect sequential two-step protonation from O_2^{2-}, followed by the desorption of H_2O_2 into the liquid phase. Alternatively, O_2 can be formed through a sequential process, in which O_2^{2-} captures successively two holes, or OOH^- undergoes a sequential hole trapping and a deprotonation process. The corresponding energies for all reactions are given in Table S2.

To determine which pathways are the kinetically most favored and which products are mainly formed in the presence of multiple competing reactions, we carried out the micro-kinetic simulations under the experimental conditions (see details in the Supporting Information). The reaction rates for O_2 and H_2O_2 generation at two interfaces are depicted in Figure 1a. At the water/b-TiO_2(210) interface, H_2O_2 is the primary product, exhibiting a formation rate approximately 3.7 times higher than that of O_2. At the water/r-TiO_2(110) interface, on the other hand, O_2 is the predominant product, accounting for 100%, which is consistent with the experimental results.[7,47,48] In addition, the overall photocatalytic activity and selectivity affected by the hole concentration is also analyzed (see details in Figure S3).

Figure 1b illustrates that the $OH^{\bullet}/OH^{\bullet}$ coupling pathway is the fastest rate for O−O bond formation at the water/b-TiO_2(210) interface, resulting in the formation of H_2O_2 primarily. This finding is attributed to the low coupling barrier of 0.38 eV. By contrast, the alternative O−O bond formation rates, involving $O^{\bullet-}/O^{\bullet-}$ coupling, $O^{\bullet-}/OH^{\bullet}$ coupling, and WNA, exhibit comparably slower kinetics, with relatively higher barriers (0.59, 0.57, and 0.64 eV, respectively). It should be noted that the coverage of surface $O^{\bullet-}$ species ($\theta(O^{\bullet-})$) is higher than $\theta(OH^{\bullet})$ due to the inherent exothermicity of $OH^{\bullet}$ deprotonation process (−0.43 eV), and the $O^{\bullet-}/O^{\bullet-}$ coupling rate is the second fastest O−O bond formation pathway. Nevertheless, the resulting O_2^{2-} species encounter difficulties in their protonation to form OOH^- with a barrier exceeding 1 eV, while the generated OOH^- species are inclined to deprotonate (Figure S4). Consequently, the peroxide intermediates are finally converted into O_2 rather than H_2O_2 (Table S3 and Table S4). These results show clearly that H_2O_2 is produced via the $OH^{\bullet}/OH^{\bullet}$ coupling pathway, while O_2 is generated via the $O^{\bullet-}/O^{\bullet-}$ coupling pathway at the water/b-TiO_2(210) interface, as shown in Figure 1c.

At the water/r-TiO_2(110) interface (Figure 1b), the surface $O^{\bullet-}$ is found to serve as the predominant precursor of peroxide intermediates, with a coverage approximately 5 orders of magnitude higher than $\theta(OH^{\bullet})$. This result is attributed to the relatively low barrier (0.41 eV) and exothermicity (−0.53 eV) in the $OH^{\bullet}$ deprotonation process. Importantly, the pathway of $O^{\bullet-}$ coupling with another $O^{\bullet-}$ is the fastest route for O−O

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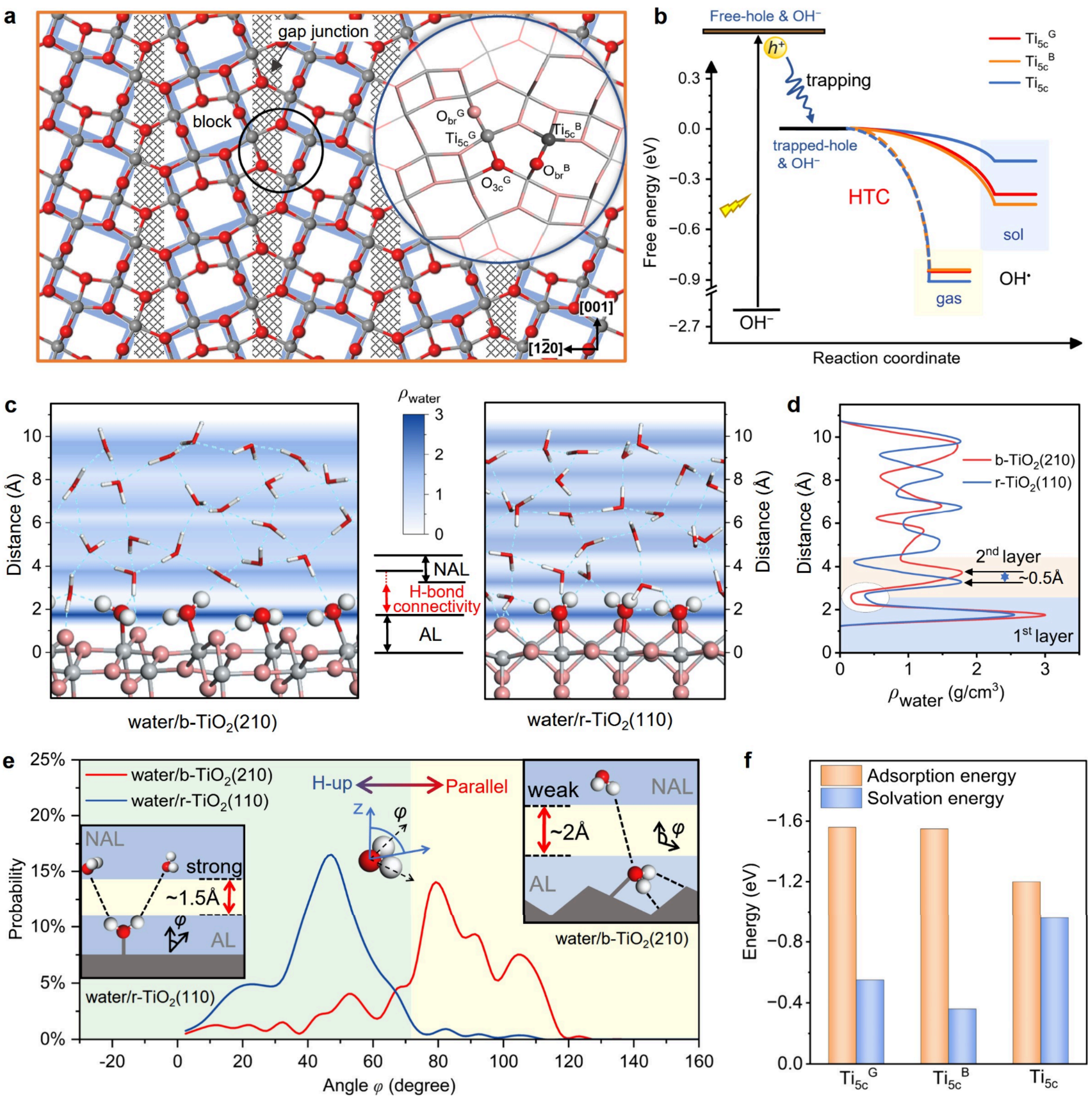

Figure 2. Formation of $OH^\bullet$ radicals at the liquid/solid interface. (a) Top view of the b-TiO_2(210) surface, where rectangles in blue and shadows in gray specify the building blocks and gap junction, respectively. Light gray and dark gray: Ti_{5c}^{G} and Ti_{5c}^{B}, respectively. (b) Scheme for the energy level of hole trapping capacity (HTC) of OH^- on b-TiO_2(210) and r-TiO_2(110) surfaces at the solvent environment (sol) and gas conditions (gas), respectively. (c) Representative snapshots of the AIMD simulation for the water/b-TiO_2(210) and water/r-TiO_2(110) systems, where the densities of water (ρ_{water}) are shown with color maps in the background (unit: g/cm^3). (d) Water density distributions along the direction perpendicular to the two interfaces. (e) Angle distributions (φ) of H_2O_{ad} at the adsorption layer (AL) within 3 Å from the catalyst surfaces, where the changes of water distribution at near adsorption layer (NAL) are inserted. (f) Adsorption energy and solvation energy of H_2O_{ad} at different Ti_{5c} sites on b-TiO_2(210) and r-TiO_2(110) surfaces. Note that the results in (d)−(f) are obtained from the equilibrium structures in the AIMD trajectories (duration from 5 to 9 ps in Figure S1).

bond formation with a barrier of 0.25 eV, which is lower than those in the other O−O bond formation pathways. The produced O_2^{2-} species undergo a strongly exothermic two-hole sequential oxidation process, yielding O_2, which is 3 orders of magnitude faster than O_2^{2-} protonation (Figure S5, Table S3 and S4). Consequently, under the experimental conditions, it predominantly forms O_2 via the $O^{\bullet-}/O^{\bullet-}$ coupling at the water/r-TiO_2(110) interface, and the production of H_2O_2 is negligible (Figure 1c).

Major Obstacle to the Selectivity of WOR. To shed more light on the main obstacle that controls the selectivity of WOR, the formation rates of H_2O_2 and O_2 were compared by analyzing the rate equation of each elementary step under experimental conditions (pH = 7; T = 298 K) (see the detailed derivation in the Supporting Information).

$$\frac{r_{H_2O_2}}{r_{O_2}} = \frac{k_{6+} \times \theta(OH^\bullet)^2}{k_{7+} \times \theta(O^{\bullet-})^2} \tag{5}$$

where k_{6+} and k_{7+} correspond to the forward rate constants of step 6 ($OH^\bullet/OH^\bullet$ coupling) and step 7 ($O^{\bullet-}/O^{\bullet-}$ coupling), respectively, which are determined by the reaction barriers. According to the equation, the main obstacle that controls the selectivity of WOR can be divided into two factors: (i) the values of $\theta(OH^\bullet)$ and $\theta(O^{\bullet-})$ and (ii) the barriers associated with $OH^\bullet/OH^\bullet$ coupling and $O^{\bullet-}/O^{\bullet-}$ coupling

Formation of $OH^\bullet$ Radicals at the Liquid/Solid Interface. Under photoirradiation, the hole can trap at OH^- on the Ti_{5c} site and oxidize it into an $OH^\bullet$ radical ($OH^- + h^+ \rightarrow OH^\bullet$). To evaluate the difficulty of $OH^\bullet$ formation at the water/catalyst interface, we calculated the hole trapping capacity (HTC) of $OH^•$ in water surrounding relative to a self-trapped hole in the bulk phase (see structures in Figure S6). The geometric structure of the b-TiO_2(210) surface is illustrated in Figure 2a, which comprises closely packed blocks arranged in a herringbone-like structure with a narrow gap junction between rows along the [001] direction. There are two types of 5-fold Ti sites suitable for $OH^•$ generation, namely Ti_{5c}^G at the gap junction and Ti_{5c}^B within the blocks (see Figure 2a). On the r-TiO_2(110) surface, there is only one type of 5-fold Ti sites (Ti_{5c}) for the reaction. Figure 2b demonstrates that the HTC of $OH^•$ at the water/b-TiO_2(210) interface is −0.39 eV at Ti_{5c}^G (−0.43 eV at Ti_{5c}^B), which is stronger than that at the water/r-TiO_2(110) interface (−0.19 eV). This finding aligns with the experimental results indicating a higher production rate of $OH^•$ radicals on b-TiO_2.[18,49−52] Notably, we also studied the inherent activity for $OH^•$ generation on the two surfaces in the absence of water, which shows that both surfaces have equivalent activity for $OH^•$ generation (Figure S7). Specifically, the HTC values of $OH^•$ calculated at the gas/catalyst interface are presented in Figure 2b: −0.85 eV, −0.84 eV, and −0.91 eV for the Ti_{5c}^G, Ti_{5c}^B sites on b-TiO_2(210), and the Ti_{5c} site on r-TiO_2(110), respectively. These results indicate that in the absence of a water solution, the holes in the two phases exhibit a comparable driving force for migration toward the surface OH^-. In other words, the HTC values of $OH^•$ on the aqueous interface are less exothermic than that at the gas/surface interface, suggesting that the solvation effect weakens the $OH^•$ formation. Furthermore, the HTC value at the water/b-TiO_2(210) interface is more negative than that at the water/r-TiO_2(110) interface, illustrating that the weakening effect of the solvation on b-TiO_2(210) surface is less significant, which can also be drawn from the layer-resolved density of state (DOS) analysis (see details in Figure S8). Hence, the interface microenvironment has significant impact on the formation of $OH^•$ radical on the catalyst surface with different crystal phases.

Furthermore, we explored the characteristics of the interfacial microenvironment modulated by the local surface structure. Figure 2c illustrates the representative structure for the water/b-TiO_2(210) and water/r-TiO_2(110) interfacial systems. As revealed in Figure 2d, H_2O_{ad} in the adsorption layer (AL) at the two interfaces appear at the similar position, about 1.8 Å from the surface. However, the near adsorption layer (NAL) water at the water/b-TiO_2(210) interface exhibits a larger peak distance (~3.7 Å) relative to the water/r-TiO_2(110) interface (~3.2 Å). Meanwhile, the water density of water/b-TiO_2(210) interface in the junction space between AL and NAL is lower than that of the water/r-TiO_2(110) interface. Furthermore, the distribution of water layers is further checked by larger supercell and thicker water layers, which shows the similar trend (see details in Figures S9 and S10). These results indicate that the H-bond interaction of water molecules between the AL and NAL is weakened at the water/b-TiO_2(210) interface. This is consistent with our calculations which show that H_2O_{ad} on the Ti_{5c}^G and Ti_{5c}^B sites experiences less solvation influence at the water/b-TiO_2(210) interface than at the water/r-TiO_2(110) interface (Figure 2f). Therefore, the formation of $OH^•$ at the water/b-TiO_2(210) interface is less influenced by the solvation effect, explaining why the $\theta(OH^•)$ is larger than that at the r-TiO_2(110) interface (Figure 1b), differing by about 3 orders of magnitude in our kinetic results. In addition, it may be worth noting that the directed migration of holes toward the interface, owing to being less affected by the solvation effect, will further decrease the electron−hole recombination rate. This finding aligns with the experimental result that the b-TiO_2 shows lower rate of electron−hole recombination.[18,53]

To further elucidate the factors governing the interfacial water network, we investigated the different water adsorption characteristics and the corresponding water network distribution. First, the unique block arrangement of the b-TiO_2(210) surface enhances the H_2O adsorption and exposes more active sites per unit area (Figure 2a).[16,54] As show in Figure 2f, the H_2O adsorption energy is stronger at the b-TiO_2(210) interface (−1.56 eV) compared to the water/r-TiO_2(110) interface (−1.20 eV). This increase strengthens the interaction between the AL and surface, while reducing the interaction between the AL and NAL. Second, it can be observed in Figure 2c that the majority of H_2O_{ad} at the water/b-TiO_2(210) interface prefer parallel orientations with respect to the surface, particularly with the H atom oriented toward the bridge oxygen (O_{br}) site, which can be attributed to the orthorhombic structure of b-TiO_2. This preferential orientation of H_2O_{ad} can be clearly observed in structural analyses conducted in the AIMD simulation, where the angle φ between the vector of the H_2O plane and the z-axis was measured (Figure 2e). The results show that most of H_2O_{ad} have a φ around 80° at the water/b-TiO_2(210) interface, indicating a stronger preference to orient toward the O_{br} sites (parallel to the surface). This adsorption character reduces the interaction between AL and NAL, resulting in the weak H-bond connectivity. Particularly, it is important to note that the presence of this preferential parallel orientations of H_2O_{ad} on b-TiO_2(210) surface leads to a zigzag-like spatial distribution of NAL water, thus forming a cavity between AL and NAL, as evidenced by the statistical analysis of the water distribution (Figure 3); such a cavity can accommodate and stabilize the semihydrophobic $OH^•$ radical. In contrast, H_2O_{ad} at the water/r-TiO_2(110) interface predominantly exhibits a φ in the range of 30° to 60°. This suggests that H_2O_{ad} is more likely oriented toward the upper layer of water, forming more H-bonds with the NAL water and leading to strong H-bond connectivity (Figure 2c,e).

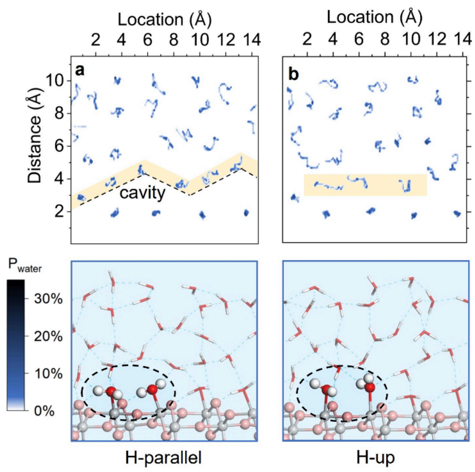

Figure 3. Statistical distribution probability of water molecules (P_{water}, projected on the x- and z-axis) at the aqueous b-TiO_2(210) interface for the H_2O_{ad} orientation in the parallel form (a) and in the H-up form (b). In each case, 500 samples were taken from their respective equilibrium stage of AIMD simulation.

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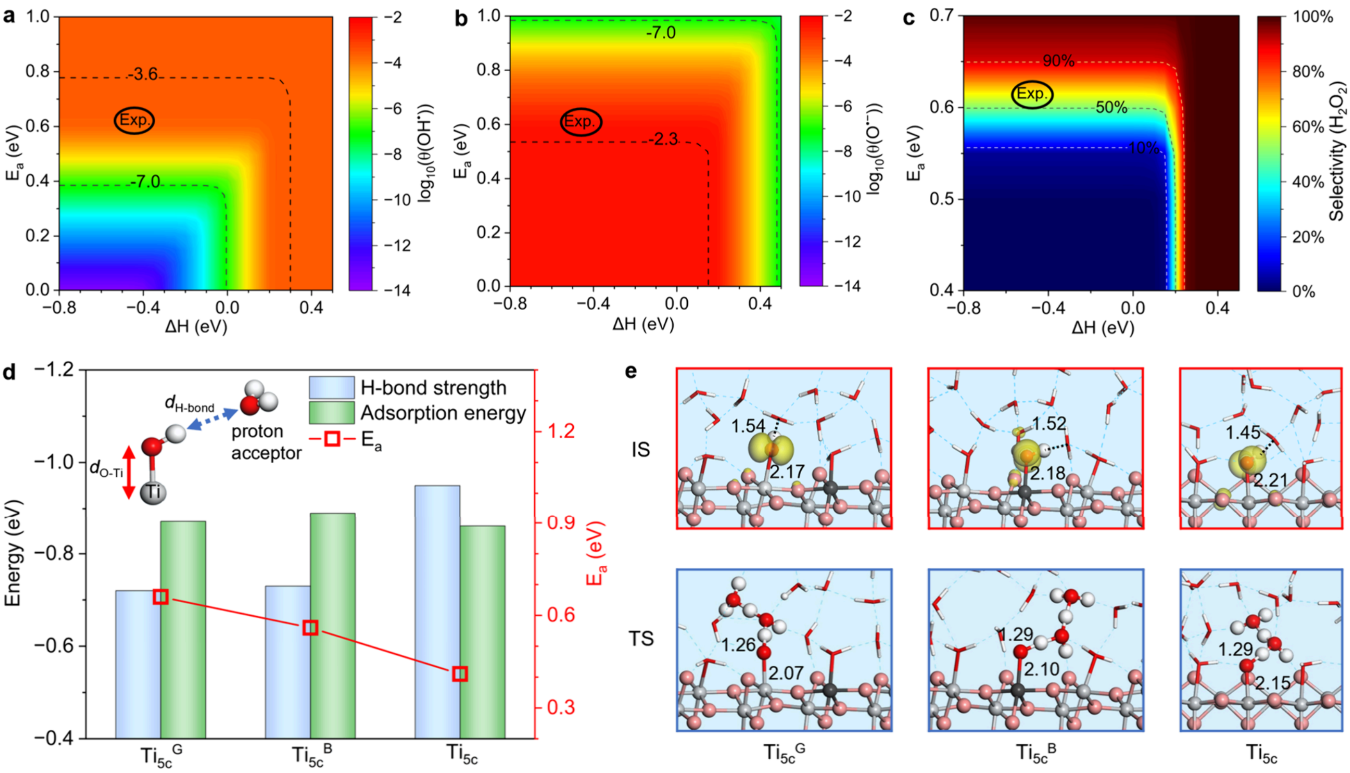

Figure 4. $OH^{\bullet}$ deprotonation at the liquid/solid interface. Contour plots of $\log_{10}(\theta)$ for $OH^{\bullet}$ (a) and $O^{\bullet-}$ intermediates (b) as a function of the reaction barrier (E_a) and enthalpy change (ΔH) of $OH^{\bullet}$ deprotonation process at the water/b-TiO_2(210) interface. (c) Contour plot of the H_2O_2 selectivity influenced by E_a and ΔH of $OH^{\bullet}$ deprotonation at the water/b-TiO_2(210) interface. (d) H-bond strength and adsorption energy of the surface adsorbed $OH^{\bullet}$ at different Ti_{5c} sites, as well as the corresponding E_a for $OH^{\bullet}$ deprotonation. (e) Spin density plots (isovalue = 0.005) of the initial state (IS) and transition state (TS) for $OH^{\bullet}$ deprotonation at the water/b-TiO_2(210) and water/r-TiO_2(110) interfaces, respectively.

Ratio of $\theta(OH^{\bullet})$ to $\theta(O^{\bullet-})$ Affected by $OH^{\bullet}$ Deprotonation. The $OH^{\bullet}$ deprotonation is a significant process that determines the ratio of $\theta(OH^{\bullet})$ to $\theta(O^{\bullet-})$. This process is essentially a surface acid−base reaction ($OH^{\bullet} \rightarrow O^{\bullet-} + H^{+}$(sol)). Our MPA-MD results show that the $OH^{\bullet}$ deprotonation is exothermic at two solid/liquid interfaces (−0.44, −0.42, and −0.53 eV at Ti_{5c}^G, Ti_{5c}^B on b-TiO_2, and Ti_{5c} on r-TiO_2, respectively), which is consistent with the experimental result that the deprotonation of adsorbed $OH^{\bullet}$ on TiO_2 under aqueous solution is an exothermic process by approximately −0.4 eV.[55,56] As shown in Figure 4a−c, our kinetic analysis demonstrates that $\theta(OH^{\bullet})$, $\theta(O^{\bullet-})$, and the selectivity of H_2O_2 are significant impacted by the $OH^{\bullet}$ deprotonation barrier. Figure 4d illustrates that the barrier of $OH^{\bullet}$ deprotonation at the water/b-TiO_2(210) interface is higher (0.56 eV at Ti_{5c}^B and 0.66 eV at Ti_{5c}^G) than that at the water/r-TiO_2(110) interface (0.41 eV), which contributes to a high ratio of $\theta(OH^{\bullet})$ to $\theta(O^{\bullet-})$. Therefore, it is essential to investigate the factor that affects the barrier of $OH^{\bullet}$ deprotonation.

During the deprotonation of $OH^{\bullet}$, the O−H bond elongates, stretching the H atom toward the nearest H_2O molecule (H_2O_{aq}, proton acceptor) in NAL (Figure 4d,e). Simultaneously, the O−Ti bond stabilizes the O atoms of $OH^{\bullet}$, forming the transition state structure (the bond length data are shown in Table S5). Therefore, in the aqueous environment, the strength of O−Ti bond and H-bond between the H atom of $OH^{\bullet}$ and the O atom of H_2O_{aq} will collectively impact on the cleavage of O−H bond. However, the O−Ti bond strength exhibits only a small variation across the three sites (Figure 4d), implying less influence than the H-bond strength (see calculation details in the Supporting Information). As shown in Figure 4d,e, with the H-bond strength decreasing in the order of $Ti_{5c} > Ti_{5c}^B > Ti_{5c}^G$, the corresponding O−H breaking barrier increases. At the water/r-TiO_2(110) interface, due to the relatively stronger H-bond strength (−0.95 eV), the H-bond distance ($d_{H\text{-bond}}$) is relatively short, ∼1.45 Å, leading to a fast deprotonation process with a barrier of 0.41 eV. As the H-bond strength of $OH^{\bullet}$ decreases at the water/b-TiO_2(210) interface (−0.72 eV at Ti_{5c}^G and −0.73 eV at Ti_{5c}^B), $d_{H\text{-bond}}$ increases (1.54 Å at Ti_{5c}^G and 1.52 Å at Ti_{5c}^B), causing the increase of deprotonation barrier to 0.66 eV (Ti_{5c}^G) and 0.56 eV (Ti_{5c}^B), respectively. Importantly, we find that the deprotonation barriers of $OH^{\bullet}$ correlate well linearly with the H-bond strength at both interfaces (Figure S11). Thus, the weak H-bond interaction between the $OH^{\bullet}$ and NAL water plays an important role in modulating the $OH^{\bullet}$ deprotonation process. These results explain why experiments showed that $OH^{\bullet}$ radicals can be detected on b-TiO_2 but not on r-TiO_2 after UV-illumination.[7,17,18,57,58]

O−O Coupling Influenced by H-Bond Network. According to eq 5, the barriers associated with O−O coupling is another crucial factor impacting the selectivity of H_2O_2. Typically, shorter Ti−Ti atomic distances are expected to promote O−O coupling with lower energy barriers.[59−61] However, the shortest Ti−Ti distance on the b-TiO_2(210) surface, 3.55 Å, is already too long and not suitable for O−O coupling, given that the typical O−O bond distance of peroxide intermediates is ∼1.5 Å. This geometric character

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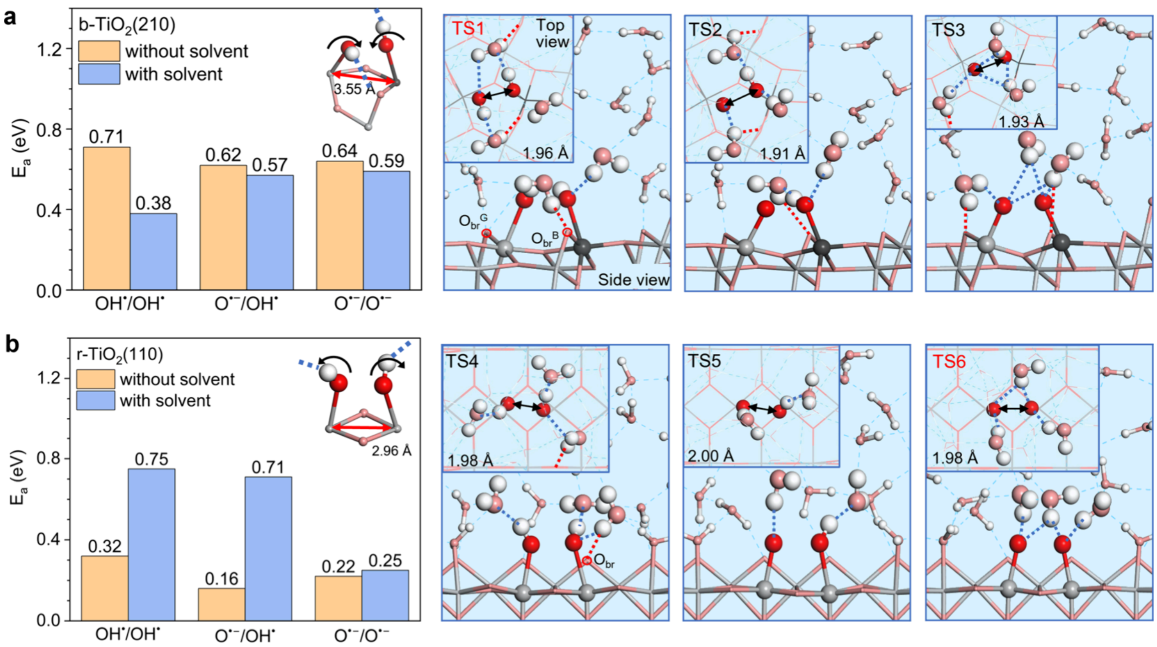

Figure 5. O−O coupling at the water/TiO_2 interface. Reaction barriers and the corresponding TS structures of three possible O−O bond formation mechanisms on the b-TiO_2(210) (a) and r-TiO_2(110) surfaces (b), respectively, in the presence and absence of solution environment. Illustrations of the influence of the Ti−Ti distance and corresponding local H-bond on oxo−oxo coupling behaviors in the two systems are inserted and marked in the histograms.

hinders O−O coupling in comparison to the 2.96 Å Ti−Ti distance on the r-TiO_2(110) surface. Thus, in the absence of water solution, the O−O coupling barrier on b-TiO_2(210) is relatively high (Figure 5a,b). However, we discovered that the H-bonds between NAL water and O_{br} play a crucial role in shaping the O−O coupling behaviors when the effects of the H-bond network were considered.

Specifically, at the water/b-TiO_2(210) interface, the $OH^•$/$OH^•$ coupling barrier is significantly reduced to 0.38 eV from 0.71 eV due to H-bond effect. The H-bond effect also reduces the barriers of the other two O−O coupling reactions to some extent (from 0.62 to 0.57 eV for $O^{•-}$/$OH^•$ and from 0.64 to 0.59 eV for $O^{•-}$/$O^{•-}$). These results are attributed to the herringbone-like structure of b-TiO_2(210) and the configuration of H_2O_{aq} in NAL, which forms the H-bond with the O_{br}^B site, making the H_2O_{aq} over the chain of the gap junction region (at the center of two coupling intermediates). In the inset (TS1) in Figure 5a, we show that H_2O_{aq} in NAL over the O_{br}^B region forms H-bonds with the adsorbed $OH^•$, with the H-bond direction increasing the possibility of the $OH^•$/$OH^•$ coupling by stabilizing the transition state, consequently facilitating H_2O_2 generation. Figure 5a depicts this effect of H-bond at the water/b-TiO_2(210) interface. Conversely, at the water/r-TiO_2(110) interface, H_2O_{aq} at the NAL forms H-bonds with $OH^•$ (or $O^{•-}$) intermediates, but the directions of H-bonds are toward the two sides of the two coupling intermediates (Figure 5b), thereby increasing the O−O bonding difficulty. For the $O^{•-}$/$O^{•-}$ coupling, the barrier increases from 0.22 to 0.25 eV. Because the effect of the H-bond network on $OH^•$ has a strong directionality, the $OH^•$/$OH^•$ coupling barrier increases to 0.75 eV from 0.32 eV and the $O^{•-}$/$OH^•$ coupling barrier increases to 0.71 eV from 0.16 eV. Therefore, the solvent effect plays a significant role in the O−O coupling and facilitates the $OH^•$/$OH^•$ coupling at the water/b-TiO_2(210) interface.

Explanation of Extended Experimental Observations. We are now at a position to provide a general picture of the role of H-bond network connectivity modulated by the local atomic structure on the selectivity of the WOR, particularly on the coverage of $OH^•$ and $OH^•$/$OH^•$ coupling barrier. First, the weak H-bond network connectivity at the water/b-TiO_2(210) interface enhances the formation of $OH^•$ radicals, as evidenced by comparing the HTC of $OH^•$ at the two interfaces under aqueous environment. This finding could explain the result of Wan et al., who demonstrated that a poor wettable interface water environment enhances the charge separation efficiency and improves the selectivity of H_2O_2.[28] This explanation could also provide a mechanistic understanding for the enhancement of photocatalytic activity by other interface environment regulations, including microdroplets[26] and hydrophobic surfaces,[29,30] which provide a relative weak H-bond connectivity reaction microenvironment. Second, at the water/b-TiO_2(210) interface, the weak H-bond network connectivity at the interface hinders the deprotonation of $OH^•$, which improves the coverage of $OH^•$ on the surface. This finding emphasizes the crucial role of $\theta(OH^•)$ in H_2O_2 production, which can explain why Cao et al. achieved 100% selectivity for H_2O_2 by modifying an abundance of hydroxyls on the b-TiO_2 surface.[4] We believe that these findings offer valuable insight into improving the selectivity of H_2O_2 production in photocatalytic processes.

■ CONCLUSION

In summary, this study presents a fascinating interface chemistry; a comprehensive understanding of the selectivity of water oxidation reaction at the water/b-TiO_2(210) and water/r-TiO_2(110) interfaces is provided. The water/b-TiO_2(210) interface exhibits a higher coverage of $OH^•$ radicals and is conducive to $OH^•$/$OH^•$ coupling, resulting in a higher selectivity of H_2O_2 in the photocatalytic WOR. In contrast, the

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water/r-TiO_2(110) interface shows significantly lower $OH^\bullet$ coverage compared to the water/b-TiO_2(210) interface, and the orientation of the H-bonds impedes O−O coupling, leading to almost no H_2O_2 generation at this interface. Such distinct selective behaviors were governed by the weak H-bond network connectivity modulated by the herringbone-like local atomic structure on b-TiO_2(210) surface, which can be summarized as follows:

(i) At the water/b-TiO_2(210) interface, the stronger water adsorption energy and distinct adsorption orientation weaken the local H-bond interaction between the water in the AL and the NAL, which leads to the formation of a weak H-bond connectivity over the b-TiO_2(210) surface.

(ii) The weak H-bond connectivity at the water/b-TiO_2(210) interface enhances the formation of semihydrophobic $OH^\bullet$ radicals and hinders the deprotonation of $OH^\bullet$, leading to an improved ratio of $\theta(OH^\bullet)$ to $\theta(O^{\bullet-})$ on the surface.

(iii) The water structure over the b-TiO_2(210) surface is conducive to $OH^\bullet/OH^\bullet$ coupling, favoring the generation of H_2O_2.

This work highlights the structure−selectivity relationship at the liquid/solid interface in photocatalysis, emphasizing that the rational manipulation of the solid/liquid interfacial structure presents a potential method for enhancing the selectivity and efficiency of photocatalytic reactions.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.3c13402.

DFT calculation details; models for water/TiO_2 interface systems; multipoint averaging molecular dynamics (MPA-MD) method; calculations of free energy and interface energy; microkinetic details; energy profile for WNA mechanism at water/r-TiO_2(110) interface; overall photocatalytic activity and selectivity affected by the hole concentration; comparation pathway between O_2^{2-} protonation and oxidation at two interfaces; structures for hole trapping at bulk oxygen and surface $OH^\bullet$ species; local DOS at two interfaces; test of water layers distribution using the larger supercell and thicker water layers models; relationship between the deprotonation barriers of $OH^\bullet$ and the interface energy, adsorption energy, and H-bond strength of $OH^\bullet$ at the interfaces; Bader charge analysis of two polymorphic surface atoms; and the bond length change during the $OH^\bullet$ deprotonation process at water/catalyst interface (PDF)

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Notes

The authors declare no competing financial interest.

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